Chemical Engineering Science 152 (2016) 26–34

Contents lists available at ScienceDirect

Chemical Engineering Science

journal homepage: www.elsevier.com/locate/ces

Mixing performance evaluation of additive manufactured milli-scale reactors

Mihret Woldemariam [a], Roman Filimonov [a,*], Tuomas Purtonen [b], Joonas Sorvari [a], Tuomas Koiranen [a], Harri Eskelinen [b]

[a] Lappeenranta University of Technology, School of Engineering Science, P.O. Box 20, FIN-53851 Lappeenranta, Finland
[b] Lappeenranta University of Technology, School of Energy Systems, P.O. Box 20, FIN-53851 Lappeenranta, Finland

HIGHLIGHTS

- New design method based on CFD and AM technology was applied to milli-reactors.
- CFD simulations were validated empirically with step responses and model reactions.
- 4-way inlet reactor with zigzag mixing channel showed the best mixing performance.

ARTICLE INFO

Article history:
Received 18 November 2015
Received in revised form
10 March 2016
Accepted 21 May 2016
Available online 24 May 2016

Keywords:
Milli-scale reactor
Step response analysis
Villermaux–Dushman parallel reaction
Computational fluid dynamics modeling
Selective laser melting
Additive manufacturing

ABSTRACT

The mixing performance of three passive milli-scale reactors with different geometries was investigated at different Reynolds numbers. The effects of design and operating characteristics such as mixing channel shape and volume flow rate were investigated. The main objective of this work was to demonstrate a process design method that uses on Computational Fluid Dynamics (CFD) for modeling and Additive Manufacturing (AM) technology for manufacture. The reactors were designed and simulated using SolidWorks and Fluent 15.0 software, respectively. Manufacturing of the devices was performed with an EOS M-series AM system. Step response experiments with distilled Millipore water and sodium hydroxide solution provided time-dependent concentration profiles. Villermaux–Dushman reaction experiments were also conducted for additional verification of CFD results and for mixing efficiency evaluation of the different geometries. Time-dependent concentration data and reaction evaluation showed that the performance of the AM-manufactured reactors matched the CFD results reasonably well. The proposed design method allows the implementation of new and innovative solutions, especially in the process design phase, for industrial scale reactor technologies. In addition, rapid implementation is another advantage due to the virtual flow design and due to the fast manufacturing which uses the same geometric file formats.

© 2016 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

1. Introduction

Mixing is one of the most important aspects of process intensification and elegant design of micromixers is key to achieving efficient production. The design of micromixers is generally regarded as a trade-off among different parameters including pressure drop, mixing time, reactor volume and velocity field (Oualha et al., 2013). Some chemical reactions demand fast mixing to enhance yield and selectivity.

Mixing of miscible fluids is classified on three length scales namely: macromixing, mesomixing and micromixing. Macromixing occurs on the scale of the vessel, mesomixing occurs on the scale of the flow eddies and micromixing occurs on the scale of molecular diffusion in stretching and folding of fluid lamellae (Johnson and Prud'homme, 2003; Bourne, 2003). For a reasonably small fluid velocity u and a hydraulic diameter D_h, typical values of convective to diffusional time scales expressed by Peclet number, uD_h/D where D is the molecular diffusion coefficient, range between 10^1 and 10^5, indicating that convection is much higher than molecular diffusion. For straight geometric micro-channels without a mixing element inside the channel, the diffusional time scale for fluid to diffuse half of the channel width is approximated by $t \sim d^2/D$ (Huysmans and Dassargues, 2005). Therefore, diffusion alone is not adequate for description of complete mixing in the shortest possible time. Innovative microchannel geometry and ideal inlet for contacting fluids will be

* Corresponding author.
E-mail address: roman.filimonov@lut.fi (R. Filimonov).

http://dx.doi.org/10.1016/j.ces.2016.05.030
0009-2509/© 2016 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

appealing factors for the microfluidic mixing to maximize the interfacial area between phases to reduce the diffusion length and enhance vortex generation (Kockmann and Roberge, 2009). Various passive micromixers have been designed to enhance mixing in microfluidic systems. Serpentine configurations are one of the popular micromixer designs. Chaotic advection generated by this type of structures enables efficient mixing by increasing the contact area between fluid streams.

A number of studies have been devoted to serpentine micromixers. Ansari and Kim (2009) conducted numerical studies on the mixing performance of a three-dimensional serpentine microchannel with L-shaped repeating units. The channel was found to be efficient in mixing at Reynolds numbers of 1, 10, 35 and 70. The mixing became more effective as the Reynolds number increased from 1 to 70. However, the results indicated that mixing performance is significantly dependent on the geometric dimensions of the channel, such as height, width, length and number of bends. The geometric parameters and Reynolds number also impact the pressure drop in the channel.

Liu et al. (2000) performed numerical and experimental investigations of mixing in a three-dimensional serpentine microchannel with C-shaped repeating units, a square-wave microchannel and a straight microchannel. Phenolphthalein and sodium hydroxide solutions were used in the empirical part of the work. It was verified by flow visualization experiments that for Reynolds number in the range from 6 to 70 a three-dimensional serpentine channel performs considerably better than the square-wave and straight channels. At a Reynolds number of 70, the straight and square-wave channels produced respectively 16 and 1.6 times less reacted phenolphthalein than the serpentine channel.

Mengeaud et al. (2002) studied both numerically and experimentally the mixing processes in zigzag microchannels. Their results showed that below a Reynolds number of approximately 80, the mixing process is ensured by diffusion only. For higher values of Reynolds number, the mixing is significantly improved by laminar recirculation processes that form around the bends of the channel. They further reported that there is a substantial improvement in the mixing efficiency when the Reynolds number is increased.

Ren and Leung (2013) investigated both numerically and experimentally the flow and mixing in a zigzag microchannel under rotation. Experimental results demonstrated that the overall mixing efficiency of a rotating zigzag channel is considerably better than that of a stationary zigzag channel and a rotating straight channel. Although the stationary zigzag channel could provide a high mixing quality, the Reynolds number needed to be sufficiently high to create a strong local centrifugal acceleration at the bends and to generate vortices. The straight channel showed the worst mixing performance.

A three-dimensional serpentine laminating micromixer with series of F-shaped mixing units was proposed by Kim et al. (2005). Two mechanisms for inducing mixing were implemented in this mixer. Chaotic advection induced by the overall three-dimensional serpentine shape of the channel was combined with a lamination mixing mechanism. The latter was implemented by a successive arrangement of F-shaped units in two layers. Numerical and experimental results showed a high level of mixing efficiency over a wide range of Reynolds numbers.

Hossain and Kim (2015) carried out a parametric study on the mixing performance of a three-dimensional serpentine split-and-recombine (SAR) microchannel with OH-shaped repeating units. They investigated the effects of geometric parameters on the mixing. The mixing performance showed a strong dependence on the ratios of the width and the height of the subchannel in the O-structure, respectively, to the width and the height of the main microchannel for Reynolds numbers greater than 15. At lower Reynolds numbers the influence of these parameters was weak.

Serpentine microchannels have thus shown good mixing capability and performance. After a certain Reynolds numbers, approximately in the 10–100 range, depending on the geometry parameters, serpentine shapes start to form vortices at the bends of the channel providing recirculation of the fluid, which plays an essential role in mixing (Mengeaud et al., 2002; Parsa et al., 2014). However, despite a large number of studies on the mixing performance of microchannels, little work has been done on the mixing performance of milli-scale reactors. The milli-scale channels provide especially higher flow rates than microchannels. Also, the blockage of channels can be prevented in large pipe structures. In this study, a Y-shaped milli-scale reactor with a straight mixing channel and Y-shaped and TT-shaped milli-scale reactors with zigzag mixing channels were designed, simulated and fabricated. The modeled results were validated by performing experimental tests. The effects of design and operating characteristics such as mixing channel shape and volume flow rate were investigated.

The main objective of this work was to demonstrate a process design method for milli-reactors which based on Computational Fluid Dynamics (CFD) modeling and Additive Manufacturing (AM) technology. The proposed design method allows the implementation of new and innovative solutions. In addition, rapid implementation is another advantage due to the virtual flow design and due to the fast manufacturing which uses the same geometric file formats as a design platform.

2. Design and fabrication

2.1. Manufacturing technologies of different channel sizes

In channel fabrication, the manufacturing process is selected based on the size and shape of the channels as well as the required surface roughness and dimensional accuracy. Mehendale et al. (2000) and Kandlikar and Grande (2003) have reviewed manufacturing technologies for mini- or compact channels. According to their research, typical methods for manufacturing micro- and milli-channels include technologies based on etching and wafer bonding processes. Suitable technologies for larger channel manufacturing include conventional milling and laser machining. Most of the aforementioned channel manufacturing processes consist of two steps: a planar open channel manufacturing and subsequent closing with a lid. The only additive manufacturing technology mentioned in their work is stereolithography, which was a commonly available technology at the time.

The applicability of AM in the construction of chemical flow reactors has been researched by a number of authors. The main difference to conventional manufacturing processes is that complex closed channels can be manufactured with AM. Dragone et al. (2013) have researched the usability of AM technology in millifluidic-device manufacturing. A material extrusion based method was used to fabricate reactionware devices out of PLA (polylactic acid). Bineli et al. (2011) have examined the manufacturing of a simple plate-type microreactor with a selective laser melting (SLM) process from aluminum alloy. Capel et al. (2013) have evaluated different AM-processes for chemical reactor manufacturing. Their conclusion was that AM-processes, including SLM, can provide greater freedom in the design of the reactors. Also, as the manufacturing rate is improved, different constructions can be more easily evaluated. Even though additive manufacturing in general is often referred to as having limitless design freedom, SLM has a number of limitations. According to Thomas (2009), overhanging structures such as rectangular channels can be difficult to fabricate. For that reason, circular or teardrop-shapes are preferred. According to Thomas, circular channels can have geometrical defects in the top section of the channel.

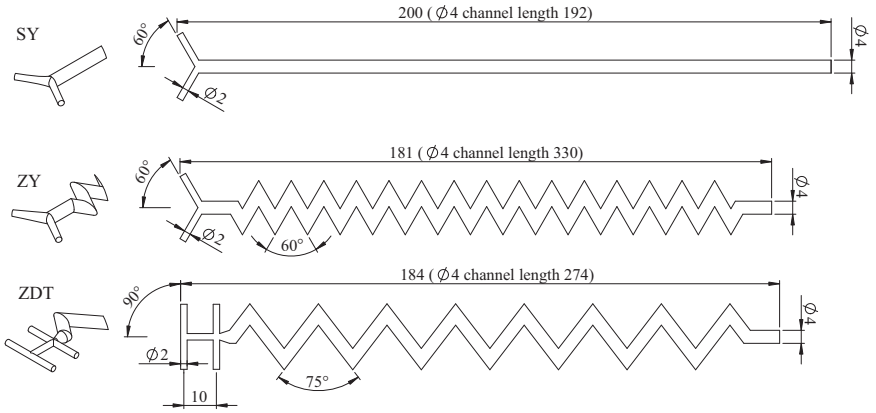


Fig. 1. The channel shapes and dimensions of the selected geometries. SY: Single Y-inlet with 2 mm diameters combining to a straight 4 mm diameter. Inlets in 60 degree corner to the combining channel; ZY: Same inlet geometry as SY, but a zigzag mixing unit; ZDT: Double T inlet with four 2 mm diameter channels attaching to a single 4 mm diameter channel. Inlets in 90 degree corner to the combining channel. Zigzag mixing unit; All the dimensions are in mm.

2.2. Design of milli-scale reactors

Fig. 1 shows the channel geometries and inlet shapes of the reactors used in this study. The manufacturing system limits the length of the mixers to approximately 200 mm and the feasible minimum diameter of the channel to 2 mm. Moderate values for the bend angles were used, since there is no optimal bend angle that provides greatly superior mixing efficiency (Ren and Leung, 2013; Li et al., 2012). The 3D-models were designed using SolidWorks software and then imported to ANSYS ICEM 14.5 software for meshing.

2.3. Fabrication of the milli-scale reactors

The reactors were designed by adding a 1 mm thick shell to the simulated geometries shown in Fig. 1. Small flat test pieces were fabricated in three different positions to emulate the surface roughness in different positions of the channel, i.e. channel top, bottom and side walls. The AISI 316L (EN 1.4404) stainless steel parts were fabricated with an EOS M-series AM system.

A diamond tip stylus profilometer was used for surface profile measurement. The measured profile was filtered with a 2CR filter to generate the roughness profile which was used for the calculation of R_a, R_z, R_{pm} and F_p values. The diameter of the channel was measured with a vernier calliper and an optical microscope. The equivalent roughness of the surface was evaluated based the equation proposed by Kandlikar (2005):

$$\varepsilon = R_{pm} + F_p \tag{1}$$

where ε is the equivalent roughness, R_{pm} is the mean leveling depth and F_p is the floor distance to the mean line. The shape evaluation was performed on five 5 mm long straight sections, which were cut from the shape estimation test piece with an abrasive saw. A backlight tool setting system and a digital SLR (single-lens reflex) camera were used to capture the shadowgraph images. The images were post processed with GIMP (GNU image manipulation program) to generate threshold images and Inkscape to attain a vector format shape of the channel. The hydraulic diameters were approximated by sketching a set of curves on top of the shadowgraph profile of the channel with SolidWorks. The same software was used to calculate the cross-sectional area and wetted perimeter of the channel. The resulting shape was used to calculate the hydraulic diameter according to the formula below:

$$D_h = \frac{4A}{P} \tag{2}$$

where D_h is the hydraulic diameter, A is the cross-sectional area of the channel and P is the wetted perimeter of the channel.

3. Numerical analysis

ANSYS Fluent 15.0 computational fluid dynamics software was utilized to perform the CFD simulations in this work. ANSYS Fluent provides comprehensive tools for fluid flow modeling and has been employed in many microfluidics studies (Mouheb et al., 2012; Wong et al., 2004; Bothe et al., 2008; Yang et al., 2015). The mixing process of two different fluids, namely pure water (bulk liquid) and a flow tracer, was simulated in the Fluent software by solving three-dimensional equations of conservation of mass and momentum and convection–diffusion. Both fluids were assumed to be independent of the gravitational field. To calculate the flow field, the steady-state Navier–Stokes equations were solved first. Then the tracer was added to the domain, and the local concentration of the tracer was obtained by solving the transient convection–diffusion equation. The respective governing equations are defined as follows:

$$\nabla \cdot \vec{u} = 0 \tag{3}$$

$$\rho \vec{u} \cdot \nabla \vec{u} = -\nabla p + \mu \nabla^2 \vec{u} \tag{4}$$

$$\frac{\partial c}{\partial t} + \nabla \cdot (\vec{u} c) = D \nabla^2 c \tag{5}$$

where $\vec{u}$ is the velocity vector, p, the pressure, ρ, the density of the fluid, μ, the dynamic viscosity of the fluid, c, the molar concentration of the species and D, the diffusion coefficient of the species. The physical properties of the fluids used in the simulations are listed in Table 1. The flow velocity at the inlets and zero static pressure at the outlet were set as the boundary conditions. No-slip boundary conditions were applied at the walls of the channels. CFD simulations and the experimental test were performed at three total flow rates, namely 12 ml/min, 18 ml/min and 30 ml/min, corresponding to Reynolds numbers of 64, 96 and 160. The Reynolds number was calculated as

$$Re = \frac{\rho u D_h}{\mu} \tag{6}$$

Table 1
Properties of fluids.

Density [kg/m^3]	Dynamic viscosity [kg/m-s]	Diffusitivity [m^2/s]
998.2	0.001003	10^{-10}

Table 2
Standard deviation of molar concentration of the tracer normalized by mean concentration value at the outlet for different mesh refinements.

Mesh setup	Total number of cells	Standard deviation
1	3,467,028	0.25
2	5,077,728	0.23
3	6,904,257	0.22

where u is the flow velocity at the inlet and D_h is the hydraulic diameter of the mixing channel. As the Re values are low, the flow was treated as laminar (Wilcox, 1998).

The computational domains were meshed with a tetrahedral grid in ANSYS ICEM 14.5 software. A mesh independence analysis was carried out in order to ensure that the simulation results are independent of the mesh size and to find the appropriate mesh resolution for the simulations. The analysis was done for the ZY-reactor. The standard deviation of the molar concentration of the tracer normalized by mean concentration value computed at the outlet surface for each mesh refinement is given in Table 2.

The difference between the computed standard deviations was 8.33% on the grids from the mesh setups 1 and 2, whereas this value was 4.44% in the setups 2 and 3. The obtained solutions indicate that for achieving sufficient accuracy with reasonable computational cost, the mesh size used in the setup 2 could be employed in the simulations. Thus, the mesh parameters used in the second case were applied for all the geometries in this study. The global maximum element size was equal to 0.4 mm, with most of the cells having all sides of about 0.1–0.2 mm length, while the hydraulic diameters of the inlet and mixing channels, accordingly, are 2 mm and 4 mm, see Fig. 1.

It is necessary to define a time step size when a transient problem is solved (Ansys, Inc, 2011). The time step size should be small enough to capture the transient behavior of the tracer properly. The initial estimate of the time step size was defined to be approximately the ratio of the average cell size to the flow velocity. However, time-dependent calculations with smaller time steps require more CPU time. To reduce the computation time of the unsteady simulations, a time step independence test was conducted. In this test, simulations were performed employing different time step sizes and the values of the standard deviation of the molar concentration of the tracer at the outlet cross-section were analyzed. To preserve accuracy and stability of the calculations, time steps were limited to a maximum of 0.008 s.

To quantify the mixing performance of the proposed reactors, the standard deviation of the mass fraction of the tracer was computed at the outlet cross-section. Then the mixing efficiency was calculated by the following formula:

$$\eta = \left(1 - \frac{1}{\bar{c}^*}\sqrt{\frac{1}{N}\sum_{i=1}^{N}(c_i - \bar{c}^*)^2}\right) \times 100\% \tag{7}$$

where N is total number of sampling nodes, c_i is the mass fraction at sample node i and $\bar{c}^*$ is the mass fraction in the case of complete mixing. A greater mixing efficiency value indicates a higher mixing quality. $\eta = 100\%$ in the case of complete mixing and $\eta = 0$ in the unmixed case. Each geometry had more than 500 sampling nodes distributed uniformly on the outlet plane in order to ensure high accuracy. Discretization of the convection term in the governing equations introduces numerical diffusion error which causes overestimation of the mixing quality in the simulation results (Mouheb et al., 2012). In order to minimize the numerical error, higher-order discretization schemes were employed in the calculations (Hardt and Schönfeld, 2003).

4. Experimental setup

4.1. Materials and methods

A number of methodologies for the evaluation of mixing performance of a reactor have been proposed, namely flow visualization, reaction and non-reactive tracer input experiments (MacMullin and Weber, 1935). The present work focuses on step response experiments and Villermaux–Dushman parallel reaction type experiment for comparison and evaluation of micromixing in the milli-structured reactors. The experiments were conducted using the following analytical grade commercially available chemicals: Sodium hydroxide, sulfuric acid and boric acid from Sigma Aldrich, potassium iodide and potassium iodate from Merck. Ultrapure Millipore water was used for all the experimental works.

4.2. Step response analysis as a tool for the evaluation of mixing efficiency

Step input experiments were conducted in a continuous flow mode using three feed streams as shown in Fig. 2. Two feed streams contained water and the third feed stream was 0.1 M solution of sodium hydroxide (tracer). First water, as the main feed stream, was pumped to the reactors using two syringe pumps from World Precession Instrument with model number AL-1000. Then one of the water lines was closed and the valve simultaneously opened for the tracer (sodium hydroxide) solution. The samples were analyzed using C3010 multi parameter pH analyzer. The concentration of the tracer was recorded every two seconds. All the experiments were done at room temperature.

4.3. Villermaux–Dushman parallel reaction as a tool for the evaluation of mixing efficiency

The Villermaux–Dushman reaction is based on two reactions that compete for an acid. The two competing reactions are given in Eqs. (8) and (9) (Guichardon and Falk, 2000):

$$H_2BO_3 + H^+ \Leftrightarrow H_3BO_3, \quad \text{quasi–instantaneous reaction} \tag{8}$$

$$IO_3^- + 5I^- + 6\,H^+ \Leftrightarrow 3I_2 + 3H_2O, \tag{9}$$

$$I_2 + I^- \Leftrightarrow I_3^- \tag{10}$$

Villermaux–Dushman reaction experiments were conducted in the milli-scale reactors using the chemicals given in Table 3 (Guichardon and Falk, 2000).

Two solutions were involved in the Villermaux–Dushman reaction. Sulfuric acid solution (Solution-1) was prepared separately. Solutions of boric acid and sodium hydroxide were prepared and mixed to form a buffer solution. Then, separately prepared solutions of potassium iodide and potassium iodate were mixed to the buffer solution. The solutions of boric acid, sodium hydroxide, potassium iodide and potassium iodate were mixed together in the same order to form a single solution (Solution-2). Iodate and

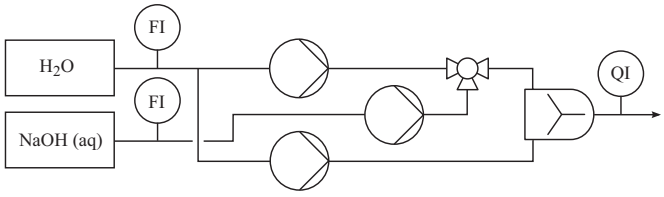


Fig. 2. Schematic drawing of experimental setup.

Table 3
Materials used in the Villermaux–Dushman reaction experiments and the respective molar concentrations. $H_2SO_4^*$ concentration is modified to suit the current reaction condition.

Material	Concentration [M]
NaOH	0.09
H_3BO_3	0.09
KI	0.032
KIO_3	0.006
H_2SO_4*	0.0026

iodide ions have a tendency to react with free hydronium ion in the solution. However, the buffer nature of the sodium hydroxide-boric acid solution prevents iodate and iodide ions from reacting with the hydronium ion from the boric acid in the solution (Commenge and Falk, 2011). Solution-1 and Solution-2 were pumped to the reactors at the same flow rate using a syringe pump from World Precession Instrument with model number AL-1000. Perfect mixing of the two solutions favors the quasi-instantaneous reaction given in Eq. (8). The concentration fluctuations due to poor mixing lead to the second reaction given in Eq. (9). The effect of poor mixing process was expressed by the formation of the triiodide ion shown in Eq. (10). The presence of the triiodide ion was detected online by a UV–vis spectrophotometer.

Samples were analyzed for the presence of triiodide ion for the parallel reaction experiment and the step response of the tracer for the non-reactive step input experiments. Analysis of the triiodide ion was done using an Agilent 8453 UV-visible spectrophotometer analyzer. Continuous online time based measurements of UV–vis spectrophotometer characterization were conducted. The UV absorption of triiodide ion at a wavelength of 353 nm was selected for the measurement. Poor mixing is manifested by the presence of a high absorbance peak of the UV curve.

5. Results and discussion

Figs. 3–5 compare normalized concentration values determined by the simulations and experiments at the outlet cross sections for different Reynolds numbers. In general, the numerical results agree well enough with those obtained experimentally. However, for all the milli-reactors studied, the concentration profile obtained from the simulation and experimental data deviates from each other with increase of the Re value. It can be noted that for Re = 160 the difference increases noticeably.

A more detailed analysis, see Table 4, was performed based on cumulative plug flow residence time distribution modeling (Fogler, 2006). According to Fogler (2006), the plug flow properties of laminar flow in pipes can be approximated by as many as an equal sized series of n continuously stirred tank reactors (CSTRs). Typically, the bigger the number of units (n) the more efficient mixing in the pipe is. The parameter values of n and τ in Table 4 are obtained by experimental data fitting for the cumulative distribution function ($F(t)$) in step change plug flow model. The unit number values indicate the best mixing for ZDT-type reactor. The CFD simulation results demonstrate the mixing for ideal geometric conditions, while the experimental n parameter values are poor due to the non-idealities in the manufactured geometry. For example, the channel surface roughness neglected in the CFD model might result in shorter residence times compared to the manufactured geometries with non-smooth wall surfaces. Therefore, the geometrical variations have an effect on the fluid distribution in the channel. Analysis of the dimensions and the roughness of the manufactured channels showed that the diameter of the main channel is 3.85 ± 0.02 mm, which is about 4% less than the nominal diameter of 4.00 mm. This shrinkage of the channel size may be a result of the poor surface roughness and non-circularity of the channels. Equivalent roughness values ε of 27 μm (R_a, 8.8 μm; R_z, 57 μm) for the channel bottom surface and 60 μm (R_a, 16.5 μm; R_z, 100 μm) for the side surfaces were determined according to Eq. (1). The top of the channel had too high roughness for the measuring equipment used. Fig. 6 shows shadowgraph images of samples A–E. Based on the literature review (Bineli et al., 2011; Capel et al., 2013), some non-circularity was expected. In addition to non-circularity, it could be seen that during the manufacturing, at apparently random positions, droplet or multiple droplets had formed on the top of the channel. The approximately 0.3 mm diameter droplets can be seen clearly on the shadowgraphs C, D and E. In sample D, there are two droplets fused together. The shadowgraph evaluations showed that the manufactured channels were not completely round, but the top section of the channels were slightly flattened, or even concave. For this reason and due to the rough surfaces of the measured samples, the hydraulic diameters of the samples were approximated. The hydraulic diameters and cross-sectional areas of the five channel shapes shown in Fig. 6 were approximated with the method described in the experimental section and calculated with Eq. (2). This resulted in an average area of 10.9 mm^2 and average D_h of 3.7 mm of the channel. Compared to the nominal values which were used in the CFD calculations, the calculated values are 13% and 7% less, respectively.

The numerical simulations provided the molar concentration profiles of the water and tracer within the flow domains of the milli-reactors. The tracer concentration data collected from the

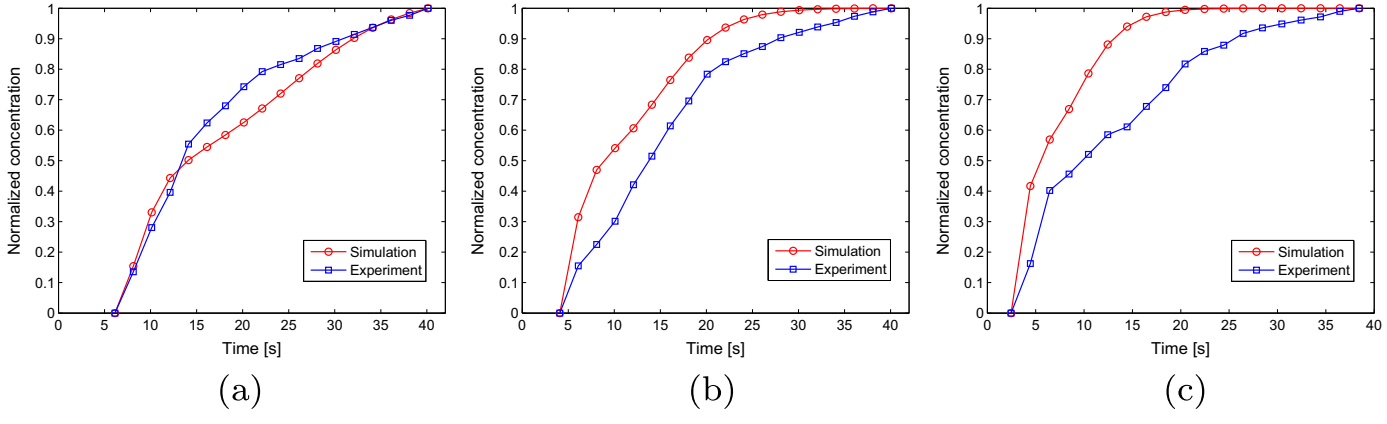


Fig. 3. Normalized concentration as a function of mixing time for the SY-reactor at the outlet at (a) Re = 64, (b) Re = 96 and (c) Re = 160. The X-axis is defined as the elapsed time from the tracer injection into the inlet channel.

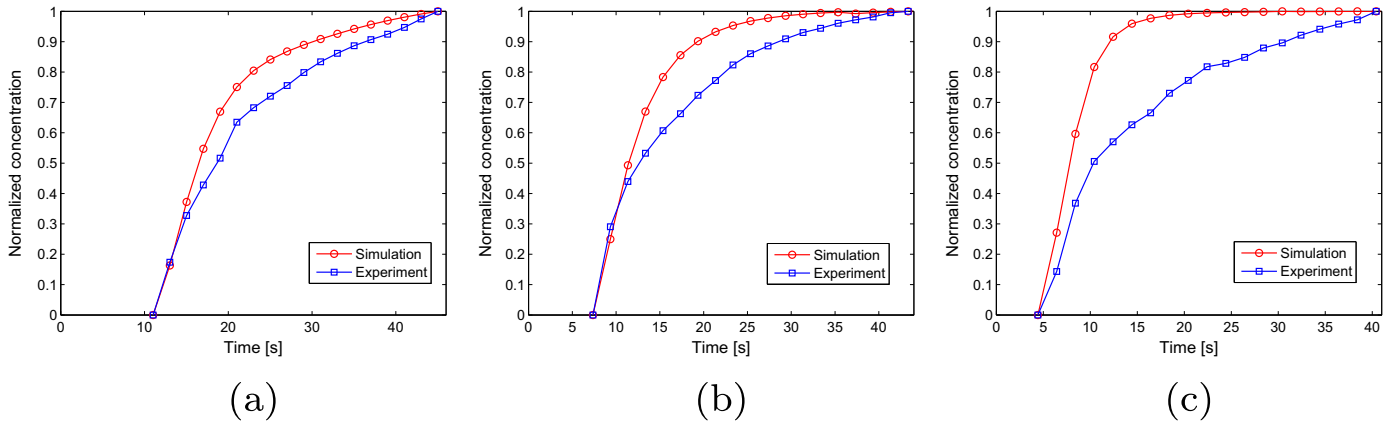


Fig. 4. Normalized concentration as a function of mixing time for the ZY-reactor at the outlet at (a) Re=64, (b) Re=96 and (c) Re=160. The *X*-axis is defined as the elapsed time from the tracer injection into the inlet channel.

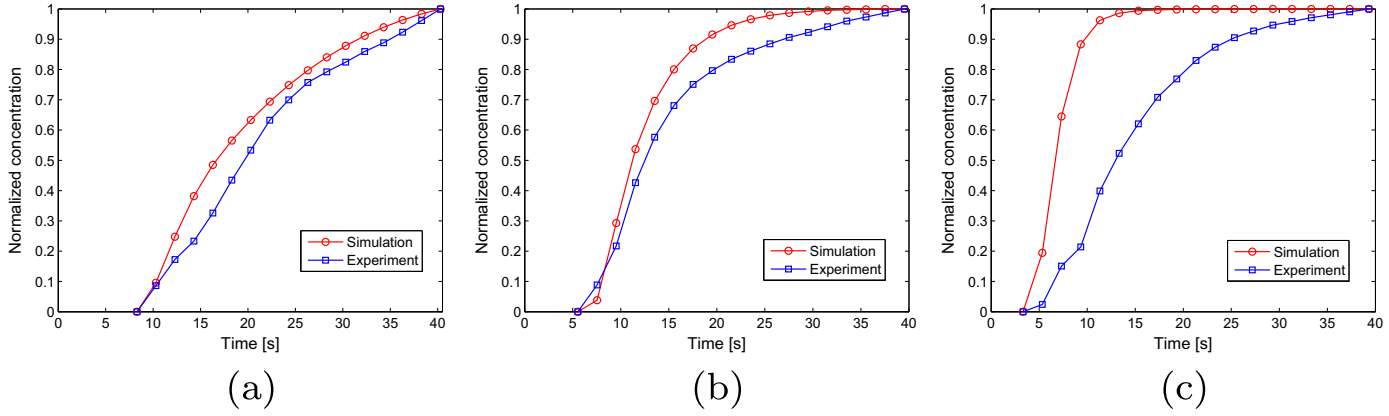


Fig. 5. Normalized concentration as a function of mixing time for the ZDT-reactor at the outlet at (a) Re=64, (b) Re=96 and (c) Re=160. The *X*-axis is defined as the elapsed time from the tracer injection into the inlet channel.

Table 4
Comparison of the number of continuously stirred tank reactor (CSTR) units (n) and mean residence time (τ) of the simulation and experimental step response concentration profiles at Reynolds numbers of 64, 96 and 160 as well as average residence times t_{res}. Times are indicated in seconds.

	Simulation			Experiment		
SY-reactor						
	Re=64	Re=96	Re=160	Re=64	Re=96	Re=160
	t_{res}=12.3	t_{res}=8.2	t_{res}=4.9	t_{res}=12.3	t_{res}=8.2	t_{res}=4.9
n	1.2	1.4	1.6	1.0	1.3	1.5
τ	14.4	6.0	5.6	14.9	13	11.3
ZY-reactor						
	Re=64	Re=96	Re=160	Re=64	Re=96	Re=160
	t_{res}=22.1	t_{res}=14.7	t_{res}=8.8	t_{res}=22.1	t_{res}=14.7	t_{res}=8.8
n	1.3	1.7	2.8	1.2	1.4	1.8
τ	9.5	6.0	4.8	12.6	10.8	10.6
ZDT-reactor						
	Re=64	Re=96	Re=160	Re=64	Re=96	Re=160
	t_{res}=16.6	t_{res}=11.1	t_{res}=6.6	t_{res}=16.6	t_{res}=11.1	t_{res}=6.6
n	2.0	2.4	3.2	1.4	1.6	2.9
τ	12.0	7.6	4.6	18.2	12.0	11.9

sampling nodes of the outlet surfaces were used to evaluate the mixing efficiency of the proposed reactors. Mixing increased with increment of the Reynolds number for all the milli-channels. In the case of the straight mixing channel, the mixing was promoted by counter-rotating vortices that arise at the intersection of the inlet channels as a result of the impact of the fluids, as shown in Fig. 7.

At higher inlet flow velocities, the effect of the vortices becomes stronger, and the mixing in the impact zone occurs more intensively and more species travel to the opposite side of the mixing channel. As shown in Fig. 8(a), the distinct interface between the streams is observed in the downstream of the impact zone, indicating no significant mixing in the channel. Thus, because of the simple geometry, the basic type of the Y-shaped milli-reactor has the lowest mixing efficiency, see Fig. 8(b). The green color on the contour plots corresponds to uniform mixing of two fluids. Therefore, in the case of well-mixed fluids, red and blue areas should not remain. Both the ZY- and ZDT-reactors showed a high level of mixing performance. For higher Reynolds number values the improvement of the mixing in the zigzag structures is associated with the flow recirculations generated at the bends, see Fig. 9. These recirculations induce a transversal component of the velocity, and thus promote the mixing. The mixing efficiency obtained is then essentially improved compared to the straight channel. To assess the effect of different geometries and flow conditions quantitatively, Table 5 summarizes the cases where the mixing efficiency approaches 90% within the 40-s simulation time interval and the corresponding elapsed times. The zigzag structures were capable to approach 90% mixing efficiency level, while the mixing efficiency values for the basic Y-type reactor were about 20%, 30% and 40% at, respectively, Re=64, 96 and 160 40 s after the tracer was introduced. The TT-type inlet allows better pre-mixing of fluids at the entrance of the mixing channel compared to the ordinary Y-shaped inlet. The improved pre-mixing induces more uniform and effective mixing in the downstream of the channel, as shown in Fig. 10. Taking also into account a shorter mixing channel length, and hence

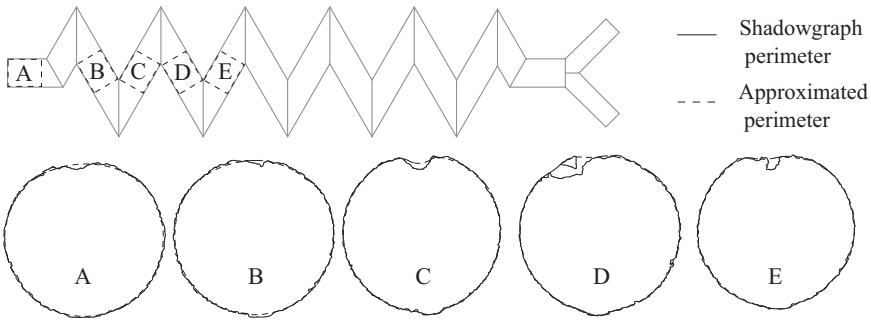
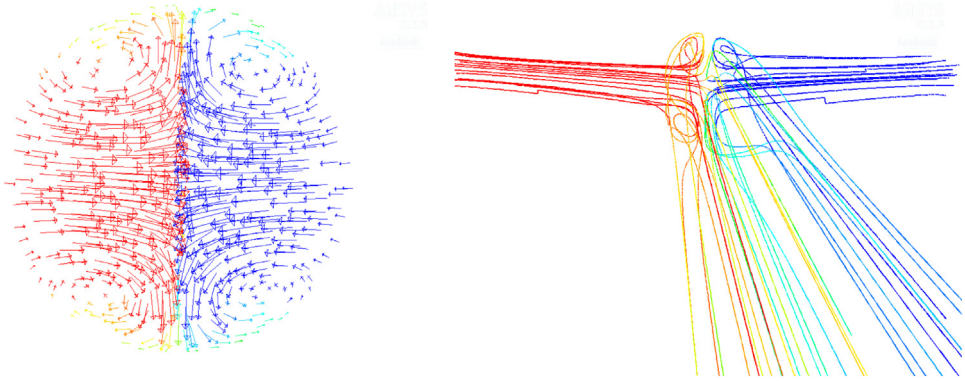


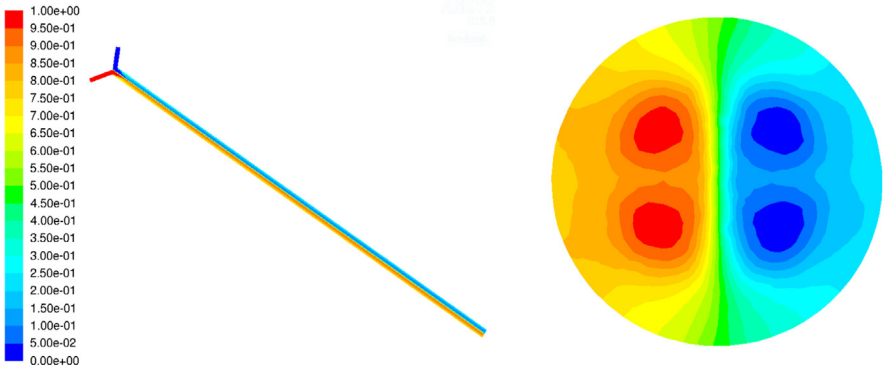
Fig. 6. Shadowgraphs and shape evaluation of the SLM-made test piece. Samples A–E were sectioned from different locations of the test piece. The shape of the channels shows random variations, which occur due to inaccuracies of the manufacturing process.



(a) Counter-rotating vortices at the impact zone of the fluids.

(b) Pathlines of species.

Fig. 7. The impact of fluids at the Y-junction of the SY-reactor. (a) Counter-rotating vortices at the impact zone of the fluids. (b) Pathlines of species.



(a) Horizontal cross-section.

(b) Outlet cross-section.

Fig. 8. Mass fraction profile of the SY-reactor at Re = 160 at 40 s state. (a) Horizontal cross-section. (b) Outlet cross-section. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

a shorter residence time, the TT-shaped reactor outperforms the zigzag reactor with Y-type inlet in terms of the time required to reach a higher mixing efficiency value.

The mixing performance of the milli-scale reactors were analyzed using online UV spectrophotometry of the triiodide ion. It is observed that the increase in Re value was associated with a decrease in UV-absorbance of the triiodide ion, which indicates improved mixing. Generally, at higher Re all the reactors showed enhanced performance, see Table 6. Significant mixing improvement was achieved at Re = 160 for the SY-reactor compared to the mixing performance at Re = 64 for the same reactor. This improvement can be directly linked to the vortices generated at the Y-junction, as illustrated in Fig. 7(a).

Of the reactors studied, the ZDT-reactor showed the best performance. However, the performance of the reactors does not only depend on Reynolds number, but also on the geometry of the channel and the feed arrangement. The results in Table 6 show the variation in mixing performance in the ZDT-reactor due to the orientation of the feed setup. In the case of ZDT1, the acid solution (Solution-1) was fed through the first T of the TT inlets as shown in Fig. 11(a). Under this setup the mixing performance of ZDT1 was shown to be slightly improved. One possible reason for the improvement is the more efficient mixing volume where the reactants are contacted with each other in the case of ZDT1. Despite possible flow variations detrimental to mixing at the different inlets of the TT-mixer, the results show that the mixing

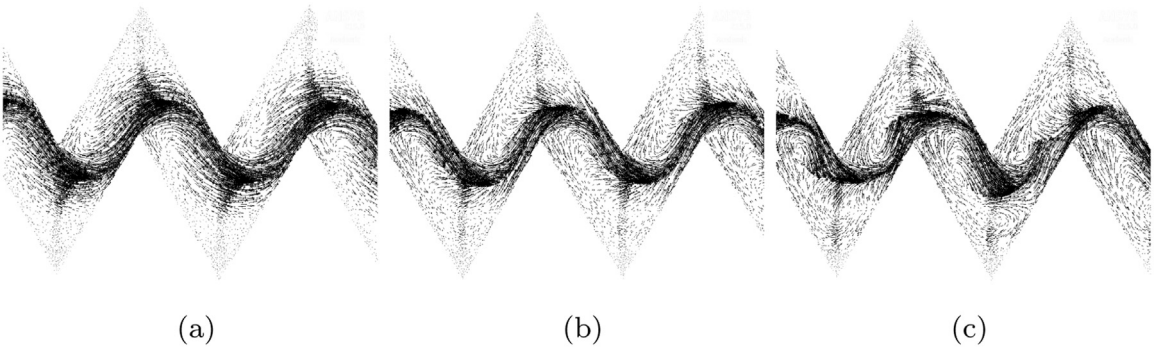


Fig. 9. Velocity vectors obtained for the zigzag configuration of the Y-shaped reactor at (a) Re=64, (b) Re=96 and (c) Re=160.

Table 5
Time taken by the reactors to achieve 90% mixing efficiency. The efficiencies were evaluated within a 40-s simulation time interval at the outlet using Eq. (7).

ZY-reactor		ZDT-reactor		
Re=96	Re=160	Re=64	Re=96	Re=160
25.5 s	17 s	40 s	20.5 s	10.5 s

Table 6
Comparison of UV–vis absorbance of the triiodide ion in the milli-scale reactors at different Re values.

Re	SY	ZY	ZDT2	ZDT1
64	1.25	0.8	0.775	0.763
96	1.03	0.78	0.774	0.759
160	0.86	0.77	0.745	0.733

performance in both the ZDT1 and ZDT2 setups was better than the other studied reactors. It can be observed that the UV absorbance experiment complements results found from the tracer experiments. The results in Tables 5 and 6 show that the ZDT outperforms the ZY milli-reactor at all Reynolds numbers considered. Repeated sets of experiment have confirmed the reproducibility of these results.

The pressure drop characteristics in the milli-scale reactors were also evaluated. Table 7 shows pressure drop values obtained from the numerical simulations for each reactor at different Reynolds numbers. The pressure drops increase with the Re value for all the geometries. The smallest pressure drops were observed in the Y-shaped milli-reactor with a straight channel. The other reactors demonstrated larger pressure drops due to the zigzag structures. At a Reynolds number of 64, the pressure drops in the zigzag configurations of the reactors are approximately the same. However, at higher Reynolds numbers, the pressure losses that occur in the ZY-reactor are noticeably higher compared to the ZDT-reactor. This could be explained by sharper bends of the zigzag mixing channel of the ZY-reactor. Besides, the stronger curvature allowed to fit a larger number of bends for the given length of the mixing channel, thus enhancing the streamwise length of the channel, see Fig. 1. The latter also results in the pressure drop increase.

6. Conclusions

The mixing performance of three passive milli-scale reactors with different geometries was investigated at different Reynolds numbers. The reactors were designed and simulated using SolidWorks and Fluent 15.0 software respectively. Manufacturing of the devices was performed with an EOS M-series AM system. Experiments with distilled Millipore water and a solution of sodium hydroxide provided the concentration profiles at outlets. The numerical results matched reasonably well the experimental data based on time dependent concentration data. The simulations were found to be consistent with the findings of UV–vis experiments as regards the mixing efficiency of the studied reactor types. For all the geometries, the mixing performance improved with increasing Reynolds number. The Y-shaped milli-reactor with a straight channel showed the least mixing effectiveness, because of its simple geometry, while the zigzag structures demonstrated significantly higher performance due to the flow recirculations that form at the channel bends. Having a moderate pressure drop, the mixing performance of the TT-shaped configuration was shown to be the highest among the considered reactors.

The SLM AM process was demonstrated to be capable of manufacturing closed milli-scale channels. The possibility to build up a geometry of various complexity that is based on the STL-file format and made directly from laser sintered metallic material

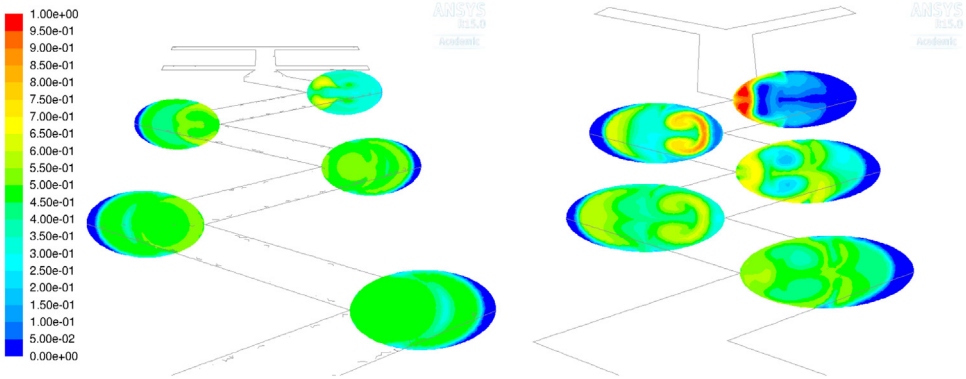


Fig. 10. Mass fraction profiles at the bends of the ZDT- (left) and ZY- (right) reactors at Re=160.

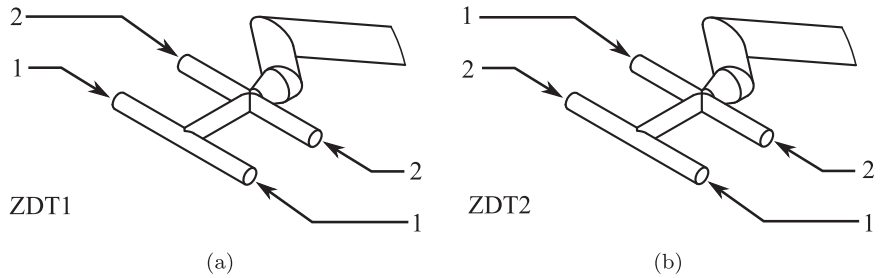


Fig. 11. Different feed setup of the ZDT milli-reactor. (a) ZDT1: Solution-1 (acid solution) was fed through inlet 1 on the first T and Solution-2 through inlet 2 on the second T. (b) ZDT2: Solution-1 was fed through 1 and Solution-2 was fed through 2 in an alternating way as shown.

Table 7
Pressure drops in the milli-scale reactors at different Re values. The pressure drop values are indicated in pascals.

Re	SY	ZY	ZDT
64	8.5	21.0	20.1
96	14.6	56.6	37.8
160	24.2	142.6	88.1

could offer greater freedom in the design of the reactors, especially in cases where the channel manufacturing would be impossible by casting, milling or drilling. However, additive manufactured flow reactors have a challenge in maintaining the accuracy of geometrical dimensions and providing smoother surfaces. Nevertheless, using CFD modeling and AM technology, the demonstrated process design method allows the implementation of new and innovative solutions for industrial scale reactor technologies, and, due to the possibility of simultaneous fabrication of several geometries at one building platform and a small number of experiments required, time savings in the presented design cycle can be remarkable when the first versions are released.

Acknowledgments

This work was supported by the Academy of Finland (Projects No. 268565 and 269896). The authors also wish to acknowledge CSC - IT Center for Science, Finland, for the provided computational resources.

References

Ansari, M.A., Kim, K.-Y., 2009. Parametric study on mixing of two fluids in a three-dimensional serpentine microchannel. Chem. Eng. J. 146 (3), 439–448.
Ansys, Inc, 2011. Canonsburg, PA, ANSYS FLUENT User's Guide.
Bineli, A.R.R., Peres, A.P.G., Jardini, A.L., Maciel Filho, R., 2011. Direct metal laser sintering (DMLS): Technology for design and construction of microreactors, in: 6° CONGRESS OBRASILEIRO DE ENGENHARIA DE FABRICA ÇÃO.
Bothe, D., Stemich, C., Warnecke, H.-J., 2008. Computation of scales and quality of mixing in a T-shaped microreactor. Comput. Chem. Eng. 32 (1), 108–114.
Bourne, J.R., 2003. Mixing and the selectivity of chemical reactions. Org. Process. Res. Dev. 7 (4), 471–508.
Capel, A.J., Edmondson, S., Christie, S.D., Goodridge, R.D., Bibb, R.J., Thurstans, M., 2013. Design and additive manufacture for flow chemistry. Lab Chip 13 (23), 4583–4590.
Commenge, J.-M., Falk, L., 2011. Villermaux-dushman protocol for experimental characterization of micromixers. Chem. Eng. Process. Process Intensif. 50 (10), 979–990.
Dragone, V., Sans, V., Rosnes, M.H., Kitson, P.J., Cronin, L., 2013. 3d-printed devices for continuous-flow organic chemistry. Beilstein J. Org. Chem. 9 (1), 951–959.
Fogler, H.S., 2006. Elements of Chemical Reaction Engineering. Prentice Hall PTR, Upper Saddle River, NJ.
Guichardon, P., Falk, L., 2000. Characterisation of micromixing efficiency by the iodide-iodate reaction system. Part I: experimental procedure. Chem. Eng. Sci. 55 (19), 4233–4243.
Hardt, S., Schönfeld, F., 2003. Laminar mixing in different interdigital micromixers: II. Numerical simulations. AIChE Journal. 49 (3), 578–584.
Hossain, S., Kim, K.-Y., 2015. Mixing analysis in a three-dimensional serpentine split-and-recombine micromixer. Chem. Eng. Res. Des. 100, 95–103.
Huysmans, M., Dassargues, A., 2005. Review of the use of Péclet numbers to determine the relative importance of advection and diffusion in low permeability environments. Hydrogeol. Journal. 13 (5–6), 895–904.
Johnson, B.K., Prud'homme, R.K., 2003. Chemical processing and micromixing in confined impinging jets. AIChE J. 49 (9), 2264–2282.
Kandlikar, S., 2005. Single-phase liquid flow in minichannels and microchannels. In: Kandlikar, S., Garimella, S., Li, D., Colin, S., King, M.R. (Eds.), Heat Transfer and Fluid Flow in Minichannels and Microchannels. Elsevier, Netherland.
Kandlikar, S.G., Grande, W.J., 2003. Evolution of microchannel flow passages-thermohydraulic performance and fabrication technology. Heat. Transf. Eng. 24 (1), 3–17.
Kim, D.S., Lee, S.H., Kwon, T.H., Ahn, C.H., 2005. A serpentine laminating micromixer combining splitting/recombination and advection. Lab Chip 5 (7), 739–747.
Kockmann, N., Roberge, D.M., 2009. Harsh reaction conditions in continuous-flow microreactors for pharmaceutical production. Chem. Eng. Technol. 32 (11), 1682–1694.
Li, Y., Zhang, D., Feng, X., Xu, Y., Liu, B.-F., 2012. A microsecond microfluidic mixer for characterizing fast biochemical reactions. Talanta 88, 175–180.
Liu, R.H., Stremler, M., Sharp, K.V., Olsen, M.G., Santiago, J.G., Adrian, R.J., Aref, H., Beebe, D.J., 2000. Passive mixing in a three-dimensional serpentine microchannel. Microelectromech. Syst. J. 9 (2), 190–197.
MacMullin, R., Weber, M., 1935. The theory of short-circuiting in continuous-flow mixing vessels in series and the kinetics of chemical reactions in such systems. Trans. Am. Inst. Chem. Eng. 31 (2), 409–458.
Mehendale, S.S., Jacobi, A.M., Shah, R.K., 2000. Fluid flow and heat transfer at micro- and meso-scales with application to heat exchanger design. Appl. Mech. Rev. 53 (7), 175–193.
Mengeaud, V., Josserand, J., Girault, H.H., 2002. Mixing processes in a zigzag microchannel: finite element simulations and optical study. Anal. Chem. 74 (16), 4279–4286.
Mouheb, N.A., Malsch, D., Montillet, A., Solliec, C., Henkel, T., 2012. Numerical and experimental investigations of mixing in T-shaped and cross-shaped micromixers. Chem. Eng. Sci. 68 (1), 278–289.
Oualha, K., Bouaziz, S., Jia, Z., Haddar, M., Kanaev, A., Amar, M.B., 2013. Design of T-mixer with non-coaxial inputs for preparation of monodispersed nanoparticles, in: SFGP2013.
Parsa, M.K., Hormozi, F., Jafari, D., 2014. Mixing enhancement in a passive micromixer with convergent-divergent sinusoidal microchannels and different ratio of amplitude to wave length. Comput. Fluids 105, 82–90.
Ren, Y., Leung, W.W.-F., 2013. Flow and mixing in rotating zigzag microchannel. Chem. Eng. Journal. 215, 561–578.
Thomas, D., 2009. The development of design rules for selective laser melting, (Ph. D. thesis). University of Wales, Wales.
Wilcox, D.C., 1998. Turbulence modeling for CFD. DCW industries La Canada, CA.
Wong, S.H., Ward, M.C.L., Wharton, C.W., 2004. Micro T-mixer as a rapid mixing micromixer. Sens. Actuators B: Chem. 100 (3), 359–379.
Yang, A.-S., Chuang, F.-C., Chen, C.-K., Lee, M.-H., Chen, S.-W., Su, T.-L., Yang, Y.-C., 2015. A high-performance micromixer using three-dimensional tesla structures for bio-applications. Chem. Eng. J. 263, 444–451.